3.4 Practice SOLUTIONS

1. **Multiple Choice:** Select the best answer for each of the following

$\cos\left(2\cdot\frac{\theta}{2}\right) = 1-2\sin^2\left(\frac{\theta}{2}\right)$
$\frac{2}{3} = 1-2\sin^2\left(\frac{\theta}{2}\right)$
$\frac{1}{6} = \sin^2\left(\frac{\theta}{2}\right)$
$\sqrt{\frac{1}{6}} = \sin\left(\frac{\theta}{2}\right)$
$\frac{1}{\sqrt{6}} = \sin\left(\frac{\theta}{2}\right)$
$\frac{\sqrt{6}}{6} = \sin\left(\frac{\theta}{2}\right)$

i. Given $\cos(\theta) = \frac{2}{3}$ in the first quadrant, the value of $\sin\left(\frac{\theta}{2}\right)$ is: **C**

a. $\frac{\sqrt{3}}{2}$ b. $\frac{\sqrt{6}}{2}$ c. $\frac{\sqrt{6}}{6}$ d. $\frac{\sqrt{3}}{6}$

ii. $2\sin\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{8}\right)$ is equivalent to : **C**

↳ $\sin\left(2\cdot\frac{\pi}{8}\right) = \sin\left(\frac{\pi}{4}\right)$

a. $\sin\left(\frac{\pi}{8}\right)$ b. $\cos\left(\frac{\pi}{8}\right)$ c. $\sin\left(\frac{\pi}{4}\right)$ d. $\cos\left(\frac{\pi}{4}\right)$

iii. The exact value of $1-2\sin^2\left(\frac{\pi}{8}\right)$ is: **B**

↳ $\cos\left(2\cdot\frac{\pi}{8}\right) = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

a. $-\sqrt{2}$ b. $\frac{\sqrt{2}}{2}$ c. $-\frac{\sqrt{2}}{2}$ d. $\sqrt{2}$

2. Express as a single sine or cosine function.

a) $10\sin(x)\cos(x)$

$= 5(2\sin x\cos x)$

$= 5\sin 2x$

b) $1-2\sin^2\left(\frac{2\theta}{3}\right)$

$= \cos\left(2\cdot\frac{2\theta}{3}\right)$

$= \cos\left(\frac{4\theta}{3}\right)$

c) $5\sin(2x)\cos(2x)$

$= \frac{5}{2}\left(2\sin(2x)\cos(2x)\right)$

$= \frac{5}{2}\sin(4x)$

d) $2\cos^2(5\theta)-1$

$= \cos(10\theta)$

3. Simplify each expression. State any restrictions on the variable in the domain $[0,2\pi]$

a) $\frac{\sin(2a)}{\cos(a)} = \frac{2\sin(a)\cancel{\cos(a)}}{\cancel{\cos(a)}} = 2\sin a$, $\underbrace{a \neq \frac{\pi}{2}, \frac{3\pi}{2}}_{\text{where } \cos a = 0}$

b) $2\tan(a)\cos^2(a) = \frac{2\sin(a)}{\cancel{\cos(a)}}\cdot\cos^{\cancel{2}}(a)$

$= 2\sin(a)\cos(a)$

$= \sin(2a)$, $a \neq \frac{\pi}{2}, \frac{3\pi}{2}$

c) $2\sin^2(a)+\cos(2a) = 2\sin^2(a) + 2\cos^2(a) - 1$

$= 2(\sin^2 a + \cos^2 a) - 1$

$= 2(1) - 1$

$= 1$

4. Expand using a double angle formula.

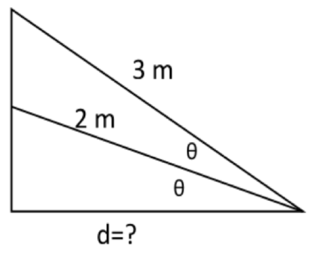
a) $3\sin(4x) = 3\left[2\sin(2x)\cos(2x)\right]$
$= 6\sin(2x)\cos(2x)$

b) $6\cos(6x) = 6\left[2\cos^2(3x) - 1\right]$
$= 12\cos^2(3x) - 6$

d) $\tan(4x) = \dfrac{2\tan(2x)}{1-\tan^2(2x)}$

e) $\cos(2x) - \dfrac{\sin(2x)}{\sin(x)} = 2\cos^2(x) - 1 - \dfrac{2\cancel{\sin(x)}\cos(x)}{\cancel{\sin(x)}}$
$= 2\cos^2(x) - 2\cos(x) - 1$

5. Two ropes (2m and 3m long) used to stabilize a pole for a volleyball net are anchored to the ground. The angle between the two ropes is equal to the angle between the ground and the lower rope. Determine the distance from the base of the pole to the point at which the ropes are anchored to the ground.



$\cos\theta = \dfrac{d}{2}$

$\cos(2\theta) = \dfrac{d}{3}$

Trig. Identity: $\cos(2\theta) = 2\cos^2\theta - 1 \Longrightarrow \dfrac{d}{3} = 2\left(\dfrac{d}{2}\right)^2 - 1$

$\dfrac{d}{3} = 2\left(\dfrac{d^2}{4}\right) - 1$

$6\left(\dfrac{d}{3}\right) = 6\left(\dfrac{d^2}{2}\right) - 1(6)$

$2d = 3d^2 - 6$

$0 = 3d^2 - 2d - 6$

$d = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a} = \dfrac{2 \pm \sqrt{4-4(3)(-6)}}{2(3)}$

$= \dfrac{2 \pm \sqrt{76}}{6}$

$= \dfrac{2 \pm 2\sqrt{19}}{6}$

$= \dfrac{1 \pm \sqrt{19}}{3}$ $\quad \dfrac{1-\sqrt{19}}{3}$ is inadmissible since $d > 0$

$\therefore$ The pole is $\dfrac{1+\sqrt{19}}{6} \doteq 1.78$ m from the base of the pole

6. a) Express $\sec(2\theta)$ in terms of $\sec(\theta)$ and $\tan(\theta)$.

$$\sec(2\theta) = \frac{1}{\cos(2\theta)} = \frac{1}{\cos^2\theta - \sin^2\theta} = \frac{1}{\cos^2\theta\left(1 - \frac{\sin^2\theta}{\cos^2\theta}\right)} = \frac{1}{\cos^2\theta} \cdot \frac{1}{1-\tan^2\theta} = \frac{\sec^2\theta}{1-\tan^2\theta}$$

b) Express $\csc(2\theta)$ in terms of $\csc(\theta)$ and $\sec(\theta)$.

$$\csc(2\theta) = \frac{1}{\sin(2\theta)} = \frac{1}{2\sin\theta\cos\theta} = \frac{1}{2} \cdot \frac{1}{\sin\theta} \cdot \frac{1}{\cos\theta} = \frac{1}{2}\csc\theta\sec\theta$$

7. Determine formulas for

a) $\sin(3\theta)$ in terms of $\sin(\theta)$

$$\begin{aligned}\sin(3\theta) &= \sin(2\theta+\theta) \\ &= \sin(2\theta)\cos(\theta) + \cos(2\theta)\sin(\theta) \\ &= [2\sin\theta\cos\theta]\cos(\theta) + [1-2\sin^2\theta]\sin(\theta) \\ &= 2\sin\theta\cos^2\theta + \sin\theta - 2\sin^3\theta \\ &= 2\sin\theta(1-\sin^2\theta) + \sin\theta - 2\sin^3\theta \\ &= 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta \\ &= 3\sin\theta - 4\sin^3\theta\end{aligned}$$

b) $\cos(3\theta)$ in terms of $\cos(\theta)$

$$\begin{aligned}\cos(3\theta) &= \cos(2\theta+\theta) \\ &= \cos(2\theta)\cos(\theta) - \sin(2\theta)\sin(\theta) \\ &= [2\cos^2\theta - 1]\cos(\theta) - [2\sin(\theta)\cos(\theta)]\sin(\theta) \\ &= 2\cos^3(\theta) - \cos(\theta) - 2\sin^2(\theta)\cos(\theta) \\ &= 2\cos^3(\theta) - \cos(\theta) - 2[1-\cos^2(\theta)]\cos(\theta) \\ &= 2\cos^3(\theta) - \cos(\theta) - 2\cos(\theta) + 2\cos^3(\theta) \\ &= 4\cos^3(\theta) - 3\cos(\theta)\end{aligned}$$

d) $\tan(3\theta)$ in terms of $\tan(\theta)$

$$\begin{aligned}\tan(2\theta+\theta) &= \frac{\tan(2\theta)+\tan(\theta)}{1-\tan(2\theta)\tan(\theta)} \\ &= \frac{\left(\frac{2\tan\theta}{1-\tan^2\theta}\right) + \tan\theta}{1 - \left(\frac{2\tan\theta}{1-\tan^2\theta}\right)\tan\theta} \\ &= \frac{\frac{2\tan\theta + \tan\theta(1-\tan^2\theta)}{1-\tan^2\theta}}{1 - \frac{2\tan^2\theta}{1-\tan^2\theta}} \\ &= \frac{3\tan\theta - \tan^3\theta}{1-\tan^2\theta} \div \frac{1-\tan^2\theta - 2\tan^2\theta}{1-\tan^2\theta} \\ &= \frac{3\tan\theta - \tan^3\theta}{\cancel{1-\tan^2\theta}} \times \frac{\cancel{1-\tan^2\theta}}{1-3\tan^2\theta} \\ &= \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}\end{aligned}$$

8. Express $\sin(2\theta)$ and $\cos(2\theta)$ in terms of $\tan(\theta)$

$$\sin(2\theta) = 2\sin\theta\cos\theta \times \frac{\cos\theta}{\cos\theta}$$

$$= 2\left[\frac{\sin\theta}{\cos\theta}\right]\left[\cos\theta \times \cos\theta\right]$$

$$= 2\tan\theta \times \cos^2\theta$$

$$= 2\tan\theta \times \frac{1}{\frac{1}{\cos^2\theta}}$$

$$= 2\tan\theta \times \frac{1}{\frac{\sin^2\theta+\cos^2\theta}{\cos^2\theta}}$$

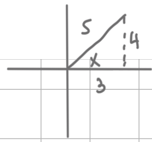
$$= 2\tan\theta \times \frac{1}{\frac{\sin^2\theta}{\cos^2\theta}+1}$$

$$= 2\tan\theta \times \frac{1}{\tan^2\theta+1}$$

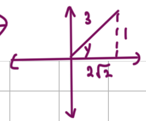
$$= \frac{2\tan\theta}{\tan^2\theta+1}$$

9. If x and y are all first quadrant angles and $\sec(x) = \frac{5}{3}$ and $\sin(y) = \frac{1}{3}$, then determine the **exact** value of $\sin(2x+y)$.

$\cos(x) = \frac{3}{5}$



$\alpha^2+3^2=5^2$
$\alpha^2=16$
$\alpha=4$



$\beta^2+1^2=3^2$
$\beta^2=8$
$\beta=\sqrt{8}$
$\beta=2\sqrt{2}$

$$\sin(2x+y) = \sin(2x)\cos(y) + \cos(2x)\sin(y)$$

$$= 2\sin(x)\cos(x)\cos(y) + \left[2\cos^2(x)-1\right]\sin(y)$$

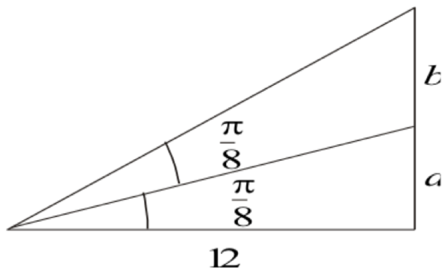
$$= 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right)\left(\frac{2\sqrt{2}}{3}\right) + \left[2\left(\frac{3}{5}\right)^2-1\right]\left(\frac{1}{3}\right)$$

$$= \frac{16\sqrt{2}}{25} + \left(-\frac{7}{25}\right)\left(\frac{1}{3}\right)$$

$$= \frac{16\sqrt{2}}{25} - \frac{7}{75}$$

$$= \frac{48\sqrt{2}-7}{75}$$

10. Find the exact value of b.



$\tan\left(\frac{\pi}{8}\right) = \frac{a}{12}$

$\tan\left(\frac{\pi}{8}+\frac{\pi}{8}\right) = \frac{b+a}{12} \Rightarrow \tan\left(\frac{\pi}{4}\right) = \frac{b+a}{12}$

$1 = \frac{b+a}{12}$

$12 = a+b$

$12-a = b$

By the identity: $\tan(2A) = \frac{2\tan A}{1-\tan^2 A}$

$\tan\left(2\cdot\frac{\pi}{8}\right) = \frac{2\tan\left(\frac{\pi}{8}\right)}{1-\tan^2\left(\frac{\pi}{8}\right)}$

$\tan\left(\frac{\pi}{4}\right) = \frac{2\left(a/12\right)}{1-\left(\frac{a}{12}\right)^2}$

$1 = \frac{a/6}{1-\frac{a^2}{144}}$

$1-\frac{a^2}{144} = \frac{a}{6}$

$144 - a^2 = 24a$

$0 = a^2+24a-144$

$a = \frac{-24 \pm \sqrt{(24)^2-4(1)(-144)}}{2(1)}$

$= \frac{-24 \pm \sqrt{1152}}{2}$

$= \frac{-24 \pm 24\sqrt{2}}{2}$

$= -12 \pm 12\sqrt{2}$

$= 12(-1 \pm \sqrt{2})$ ← $12(-1-\sqrt{2})$ is inadmissible because $a>0$

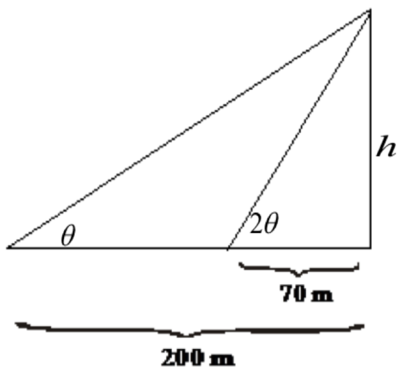
$\therefore a = 12(\sqrt{2}-1)$

$b = 12 - a = 12 - 12(\sqrt{2}-1)$

$= 12[1-\sqrt{2}+1]$

$= 12[2-\sqrt{2}]$

11. Find the value of h in the below diagram.



$\tan\theta = \frac{h}{200}$

$\tan(2\theta) = \frac{h}{70}$

By trig. identity:

$\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$

$\frac{h}{70} = \frac{2\left(h/200\right)}{1-\left(h/200\right)^2}$

$h\left[1-\left(\frac{h}{200}\right)^2\right] = 140\left(\frac{h}{200}\right)$

$h - \frac{h^3}{40000} = \frac{7h}{10}$

$40000h - h^3 = 28000h$

$0 = h^3 - 12000h$

$0 = h(h^2-12000)$

$h = 0$ (inadmissible b/c $h>0$)

$h^2 - 12000 = 0$

$h^2 = 12000$

$\boxed{h \doteq 20\sqrt{30}}$